Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

Supplementary Appendix

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LIST OF STUDY GROUP

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ADDITIONAL METHODS

Selection of Primary Endpoint

In this trial, we assessed the CDC's benchmark for type 2 diabetes risk reduction, which has been validated in previous studies as a reliable surrogate for decreased incidence of diabetes.

Both the DPP and Finnish DPP reported a significant reduction in diabetes incidence among individuals achieving 5% to 7% weight loss.[1,2] While the specific recommendation of a 4% weight loss combined with 150 minutes of physical activity is not derived from a single study, it represents a pragmatic and meaningful goal. The DPP found that each kilogram (2.2 lbs) of weight loss corresponded to a 16% reduction in diabetes risk[3]; thus, a 4% weight loss for an individual weighing 100 kg (220 lbs) equates to approximately 8.8 lbs, potentially reducing diabetes risk by around 64%.

Furthermore, individuals who met the physical activity benchmark of 150 minutes per week, but not the weight loss target, experienced a 44% decrease in diabetes risk, independent of their modest average weight loss of approximately 2.9 kg.[3] A systematic review and meta-analysis also demonstrated that engaging in 150 minutes of moderate physical activity weekly is associated with a 26% to 50% reduction in the risk of developing diabetes.[4]

Lastly, the benchmark of a 0.2% improvement in A1C at one year is supported by findings from the original DPP, which reported a 0.1% A1C improvement in the lifestyle intervention group at one year[2]; thus, a 0.2% point reduction would indicate even greater improvement in metabolic control. Similarly, in the Finnish DPP, the intervention group experienced a mean A1C reduction of 0.1% at one year, while the control group saw an increase of 0.1%.[5]

A1C Measurement

Study Member Obtaining A1C Measurements

A1C measurements were obtained by trained and certified research staff. For the A1CNow+ test kit, research staff were required to complete and successfully pass a pts Diagnostics training course and receive a certificate. For the Afinion™ 2, research staff were required to successfully complete the training provided by Johns Hopkins Medicine on the employee MyLearning platform.

Instruments Used for A1C Measurements

A1C measurements were obtained using FDA-cleared, CLIA-waived, and NGSP-certified point-of-care devices, including the Afinion™ 2 Analyzer, Siemens DCA Vantage, and A1CNow+ test kit. These tests required a fingerstick blood sample, with results available within 5 minutes. All A1C testing was conducted by a trained research coordinator.

The A1CNow+ test kit was a portable device used primarily for home study visits (PTS Diagnostics). Although FDA-approved and NGSP-certified, it exhibited a larger mean bias compared to laboratory-based measurements and other point-of-care devices (Afinion™ 2 Analyzer and Siemens DCA Vantage). Additionally, A1CNow+ was more susceptible to interference from hemoglobinopathies (e.g., sickle cell trait or disease) and rheumatoid factor.

To mitigate potential inaccuracies, a screening protocol was implemented for participants requesting home study visits:

- Participants with conditions known to interfere with the A1CNow+ test were excluded from home testing and instead scheduled for in-clinic testing using more accurate devices (Afinion™ 2 Analyzer or Siemens DCA Vantage).
- If an A1CNow+ test produced a quality control error message, the test was repeated with a second sample.
- A1CNow+ failure was defined as two consecutive quality control or error messages during the same visit.
- If an A1CNow+ result deviated significantly ($\leq$ -0.9 or $\geq$ +0.8 points) from a prior A1C measurement within the past three months (i.e., outside the 95% confidence interval for A1CNow+ compared to venous sample A1C), an alternative A1C test was offered at the clinical research unit or a Labcorp facility.

The following table summarizes the number of participants who underwent A1C testing using each device at baseline and at the 12-month follow-up:

A1C Measurement Device/Method	Baseline (n)	12-Month Follow-up (n)
Afinion™ 2 Analyzer	334	282
A1CNow+	33	30
Serum (Labcorp facility)	1	0

Additionally, some participants transitioned between different A1C testing methods between baseline and follow-up:

Participants with A1C measurements with discordant devices:

Baseline Device → 12-Month Follow-up Device	No. of Participants (% of Trial Population)
Afinion™ 2 Analyzer → A1CNow+	6 (1.6)
A1CNow+ → Afinion™ 2 Analyzer	1 (0.27)
Serum → Afinion™ 2 Analyzer	0 (0)
Afinion™ 2 Analyzer → Serum	0 (0)
Serum → A1CNow+	1 (0.27)
Total	8 (2.2)

Body Weight Measurement

Study Member Obtaining Body Weight Measurements

Performed by trained research coordinators.

Instruments Used

All body weight measurements were taken via the Tanita WB-100A Digital Medical Scale. Scales were inspected by the Johns Hopkins Clinical Engineering annually. Before taking the weight measurements, participants were asked to use the bathroom, if needed, remove footwear, accessories such as wallets or keys, and jackets.

Physical Activity Measurement

Study Member Obtaining Physical Activity Measurements

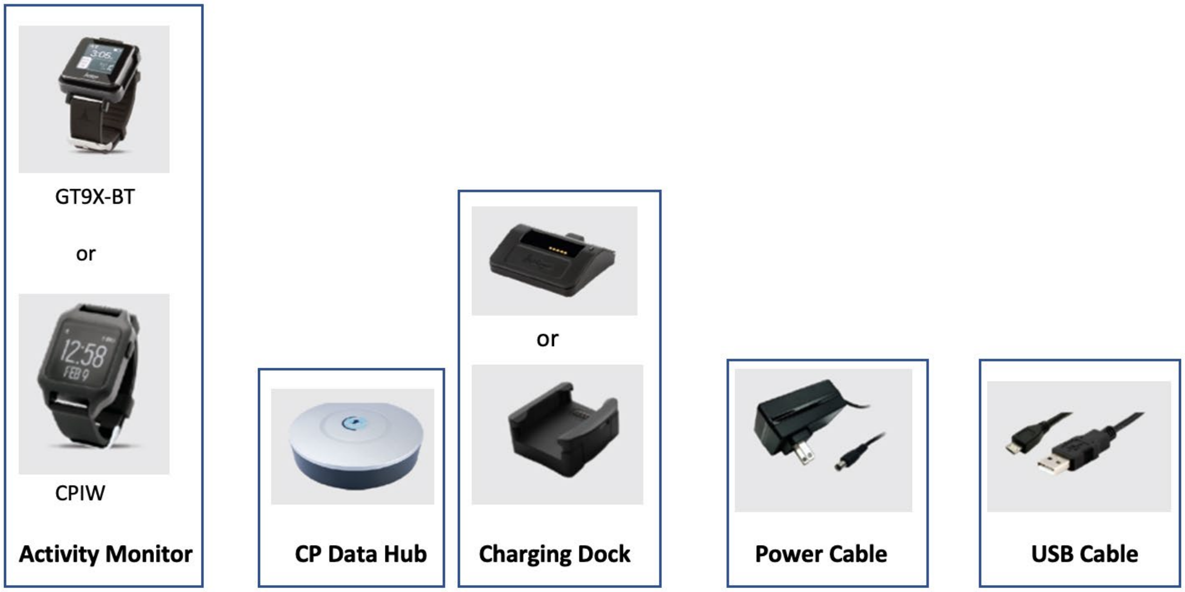
Performed by trained research coordinators.

Instruments Used

ActiGraph provided a cloud-based service platform, CenterPoint, which enabled automated and manual capture of actigraphy data from the ActiGraph monitor. At the baseline visit, participants were provided with (1) the ActiGraph activity monitor, (2) a CenterPoint (CP) data hub, (3) charging dock, (4) power cable, and (5) USB cable, along with a participant guide containing detailed instructions (pictured below). The CP data hub used a 3G cellular connection (covered by the study team) to transmit data from the activity monitor to the study team via the CenterPoint cloud software, without relying on participants' home WiFi. Using Bluetooth technology, the data hub automatically captured data when the monitor was in proximity. In addition, participants were instructed to upload raw data after each wear period by physically connecting the activity monitor to the data hub using the USB cable and charging dock. This ensured that raw high-resolution acceleration data were transmitted reliably. The study team monitored participant compliance remotely via CenterPoint, which tracked both Bluetooth-based and manual data uploads.

The instruments used to measure physical activity (MVPA) were the ActiGraph GT9X-BT Link and Centrepoint Insight Watch (CPIW) — FDA-cleared devices that can distinguish non-wear time from physical inactivity. Due to connectivity and uploading issues with some participants in the beginning of the trial, the study investigators decided to switch from the GT9X-BT to CPIW watch, to ensure the integrity and completeness of trial data. Among the 368 randomized participants, 33 were assigned to the GT9X-BT at baseline and as the switch took place, they were provided with the new CPIW and their data from the GT9X-BT was retrieved.

Both devices captured and recorded high-resolution raw acceleration data, which were converted into a variety of objective activity measures. The CPIW, which was the newer model of the ActiGraph wrist monitor, had an extended 30-day battery life, allowing for continuous wear time during the scheduled wear period. This minimized participant burden and the potential for missing data associated with frequent device charging.


GT9X-BT
or
CPIW
Activity Monitor
CP Data Hub
or
Charging Dock
Power Cable
USB Cable

Physical Activity Data Processing

Accelerometry Data Collection and Processing

Physical activity data were collected from participants using ActiGraph accelerometry devices (GT9X or CPIW) worn on the wrist. Participants were instructed to wear the devices for seven consecutive days during each scheduled visit, with each 7-day collection representing a typical week within that month. This approach allowed for sampling of physical activity while accounting for secular trends such as weather and holidays. Data collection occurred at baseline and continued monthly for a total of 12 visits.

The accelerometers recorded activity counts per minute, which were processed to identify moderate-to-vigorous physical activity (MVPA) using Montoye's cut point of 3941 counts per minute. This threshold was selected based on validated criteria for classifying MVPA in adults.[6] Activity data were collected as raw timestamped files in CSV format and imported into R for processing.

Data Preprocessing

For each file, accelerometry data were extracted, and relevant columns were renamed for consistency. The timestamp data were parsed to accurately reflect the date and time of each observation. Data were segmented into daily epochs from midnight to midnight to ensure standardized processing. Only days with at least 10 hours of wear time were considered valid for subsequent analysis. Wear and non-wear periods were identified using a 90-minute window of consecutive zero counts as the criterion for non-wear detection.[7] Valid wear days were determined by applying a threshold of at least 10 hours of wear time per day. Missing data within valid days were imputed using interpolation methods to ensure continuity in physical activity monitoring. Days without valid accelerometry were assigned zero minutes of MVPA to reflect the assumption that no moderate-to-vigorous activity was performed during those periods.

MVPA Calculation

Activity intensity was classified as MVPA if the vector magnitude of accelerometer counts per minute was equal to or greater than 3941. For each valid wear day, the total number of MVPA minutes was computed and averaged across valid days for each visit. Additionally, sustained bouts of MVPA lasting at least 10 consecutive minutes were identified using the same threshold of 3941 counts per minute. The number of these bouts and the total duration spent in them were calculated for each valid wear day. The R code that was used for this calculation is provided below.

Aggregation of MVPA Across Visits

To obtain the average MVPA minutes per week across the study period, the total MVPA minutes per week from all valid wear periods were summed and divided by the number of available wear periods (maximum of 11 post-baseline visits). This approach provided an average estimate of weekly MVPA throughout the 12-month follow-up period.

Actigraphy Compliance

If the participant did not wear their monitor for at least 5 of their 7 assigned days, they were considered noncompliant. In this case, the participant was assigned 0 minutes for physical activity for that week (See eTable18 for further details and rationale).

R Code for MVPA Calculation

The following R code was used to preprocess wrist-worn ActiGraph data and compute total and bouted MVPA using vector magnitude thresholds.

```
rm(list = ls())
# Set working directory to "code" folder
setwd('D:/OneDrive - Johns Hopkins/1 Studies/1 Ongoing/DDP trial/Accelerometry/code')

# Load packages and source bout calculation function
source('BoutsOfPA.R')
library(lubridate)
library(dplyr)
library(readr)
library(arctools)

# Define file path
file_path <- "D:/OneDrive - Johns Hopkins/1 Studies/1 Ongoing/DDP trial/Accelerometry/csv/"
files <- dir(file_path, recursive = TRUE)
files <- files[grep('.csv', files)]

# Bouts definition
bouts_min <- c(10)
bouts_max <- c(Inf)

# Summarize MVPA Activity
out_activity <- data.frame(
  id = numeric(),
  visit = numeric(),
  site = numeric(),
  tas = numeric(),
  n_days = numeric(),
  n_valid_days = numeric(),
  wear_time_on_valid_days = numeric(),
  min_spent_active_1853 = numeric(),
  min_spent_active_2185 = numeric(),
  min_spent_active_2859 = numeric(),
  min_spent_active_3941 = numeric(),
  stringsAsFactors = FALSE
)

pb <- txtProgressBar(min = 0, max = length(files), style = 3)

for (j in 1:length(files)) {
  name <- files[j]
  name.full <- paste0(file_path, name)

  id <- substr(name, 1, 3)
  visit <- substr(name, 5, 6)
  site <- substr(name, 8, 11)
```

```
  datai <- read.csv(name.full)
  colnames(datai) <- c("timestamp", "serialnumber", "axis1", "axis2", "axis3", "steps", "vectormagnitude",
"capsense")

  ac <- datai$vectormagnitude
  timestamp <- ymd_hms(datai$timestamp)
  tas <- datai$serialnumber[nrow(datai)]

  ac_mid <- midnight_to_midnight(ac, timestamp)
  wnw <- get_wear_flag(ac_mid, nonwear_0s_minimum_window = 90)
  valid.long <- get_valid_day_flag(wnw, validday_nonwear_maximum_window = 840)
  ac_imp <- impute_missing_data(ac_mid, wnw, valid.long)

  result_summary_1853 <- activity_stats(ac, timestamp)
  result_summary_2185 <- activity_stats(ac, timestamp, sedentary_thres = 2185)
  result_summary_2859 <- activity_stats(ac, timestamp, sedentary_thres = 2859)
  result_summary_3941 <- activity_stats(ac, timestamp, sedentary_thres = 3941)

  active <- data.frame(cbind(
    id, visit, site, tas,
    result_summary_1853$n_days,
    result_summary_1853$n_valid_days,
    result_summary_1853$wear_time_on_valid_days,
    result_summary_1853$time_spent_active,
    result_summary_2185$time_spent_active,
    result_summary_2859$time_spent_active,
    result_summary_3941$time_spent_active
  ))

  out_activity <- rbind(out_activity, active)
  setTxtProgressBar(pb, j)
}

close(pb)

colnames(out_activity) <- c(
  "id", "visit", "site", "tas",
  "n.weardays", "n.valid.weardays", "minutes.on.valid.weardays",
  "minutes.active.1853", "minutes.mvpa.2185",
  "minutes.lmvpa.2859", "minutes.mvpa.3941"
)

# MVPA Bouts Calculation
out_bouts <- data.frame(
  id = numeric(),
  visit = numeric(),
  site = numeric(),
  n.mvpa.10toInf_bouts.2185 = numeric(),
  minutes.mvpa.10toInf_bouts.2185 = numeric(),
```

```
  n.lmvpa.10toInf_bouts.2859 = numeric(),
  minutes.lmvpa.10toInf_bouts.2859 = numeric(),
  n.mvpa.10toInf_bouts.3941 = numeric(),
  minutes.mvpa.10toInf_bouts.3941 = numeric(),
  stringsAsFactors = FALSE
)

pb <- txtProgressBar(min = 0, max = length(files), style = 3)

for (j in 1:length(files)) {
  name <- files[j]
  name.full <- paste0(file_path, name)

  id <- substr(name, 1, 3)
  visit <- substr(name, 5, 6)
  site <- substr(name, 8, 11)

  datai <- read.csv(name.full)
  colnames(datai) <- c("timestamp", "serialnumber", "axis1", "axis2", "axis3", "steps", "vectormagnitude",
"capsense")

  ac <- datai$vectormagnitude
  timestamp <- ymd_hms(datai$timestamp)

  ac_mid <- midnight_to_midnight(ac, timestamp)
  wnw <- get_wear_flag(ac_mid, nonwear_0s_minimum_window = 90)
  valid.long <- get_valid_day_flag(wnw, validday_nonwear_maximum_window = 840)
  ac_imp <- impute_missing_data(ac_mid, wnw, valid.long)

  bouts_2185 <- BoutsOfPA(ac, valid.long, timestamp, bouts_min = bouts_min, bouts_max =
bouts_max)[, -c(1:3)]
  bouts_2859 <- BoutsOfPA(ac, valid.long, timestamp, bouts_min = bouts_min, bouts_max =
bouts_max)[, -c(1:3)]
  bouts_3941 <- BoutsOfPA(ac, valid.long, timestamp, bouts_min = bouts_min, bouts_max =
bouts_max)[, -c(1:3)]

  bouts <- data.frame(cbind(
    id, visit, site,
    bouts_2185$n.bouts_from10toInf_minutes,
    bouts_2185$time.spent.in.bouts_from10toInf_minutes,
    bouts_2859$n.bouts_from10toInf_minutes,
    bouts_2859$time.spent.in.bouts_from10toInf_minutes,
    bouts_3941$n.bouts_from10toInf_minutes,
    bouts_3941$time.spent.in.bouts_from10toInf_minutes
  ))

  out_bouts <- rbind(out_bouts, bouts)
  setTxtProgressBar(pb, j)
}
```

```
close(pb)

colnames(out_bouts) <- c(
  "id", "visit", "site",
  "n.mvpa.10toInf_bouts.2185", "minutes.mvpa.10toInf_bouts.2185",
  "n.lmvpa.10toInf_bouts.2859", "minutes.lmvpa.10toInf_bouts.2859",
  "n.mvpa.10toInf_bouts.3941", "minutes.mvpa.10toInf_bouts.3941"
)

# Merge Summary Data and Export
combine_summary <- merge(out_activity, out_bouts, by = c("id", "visit", "site"))

write.csv(
  combine_summary,
  "D:/OneDrive - Johns Hopkins/1 Studies/1 Ongoing/DDP trial/Accelerometry/result/DPP_MVPA and
bouts_07012022.csv",
  row.names = FALSE

)
```

Data Quality Processes

Clinical data (including adverse effects and concomitant medications) and laboratory data were entered into REDCap (Research Electronic Data Capture), a 21 CFR Part 11-compliant data capture system provided by Johns Hopkins University. The data system included password protection and internal quality checks, for any source of data that had impacts on the trial's outcomes for each participant. These quality measures included missing data algroups and automatic range checks for BMI, weight, A1C, Proof of A1C measurement, height, Actigraph physical activity start and end dates, and whether Actigraph physical activity was completed, to identify data that appeared inconsistent, incomplete, or inaccurate, clinical data were entered directly from the source documents.

To obtain, assess, and convert the Actigraph physical activity data, research coordinators and staff were trained to work with multiple Centrepoint test accounts, which included only test data created by staff and no actual participant data. After each month of physical activity completion, research coordinators and staff downloaded and exported the participants' data in SafeDesktop in the form of a gt3x file. During the conversions and analysis of files, research staff and coordinators carefully assessed the participants' wear period data to identify any missing wear periods. For those with missing wear periods, research staff would check the REDCap participant files for any particular notes, and CentrePoint to ensure that the data was missing due to participants not complying, rather than the staff missing the wear periods from the CentrePoint platform.

In terms of analyzing, assessing, and converting the physical activity data, several conversions and steps were required, which were completed in SafeDesktop provided by Johns Hopkins. Physical activity data was first downloaded in gt3x format and placed in SafeDesktop, where it was then converted to CSV files. CSV files were then converted to Moderate to Vigorous Physical Data (MVPA) per day in a STATA .dta file. The PI and research staff trained in STATA used this converted data and data exported from REDCap to assess physical activity outcomes and completion at 6-months and 12-months. To ensure quality of outcome data, the STATA log file that demonstrated the outcomes after coding was also uploaded to REDCap, where research coordinators would run the log file for each participant.

Research staff and coordinators the Reports feature in REDCap to create different reports for analysis purposes and identifying missing or incorrect values for different variables and fields. Reports were also used to summarize overall data and increase the efficiency of the data analysis, management, and quality control process.

Randomization

Since a higher value of A1C at baseline represented the strongest risk factor for progression to incident diabetes and demand for glucose-lowering medications, we used a block stratified randomization approach by baseline A1C value and recruitment site to ensure that groups were balanced.

This trial was conducted in an open-label manner to ensure pragmatic integrity. The randomization, and collection of participant data (e.g., weight, height) were completed by research coordinators to ensure efficiency. This also ensured efficient communication between research staff and local DPPs and prevented challenges that would've arisen if research staff were also blinded to treatment. The trial's biostatistician was also unmasked to conduct required reporting to the DSMB.

Due to the Human-DPPs rolling basis enrollment, based on amount of interested participants and instructor availability, research staff were responsible for ensuring that there were classes starting within 30 days of randomization for the participants assigned to the Human-DPP group. If this was not the case, randomization for the period was halted until at least one program start date became available within 30 days.

Study Team’s Interactions with Participants

Research coordinators and staff purposefully did not actively engage with the study participants to maintain the real-world and pragmatic replication aspect of the trial.

Overall, there were 3 in-person interactions and visits that the participants had with research staff. These visits took place at Baseline, 6-Month, and 12-Month time points of the trial, where laboratory and survey data collected from the participants.

To further eliminate unnecessary contact with the participants, any reminders regarding upcoming study visits, upcoming actigraph wear period, or end of wear period were automated based on the study schedule of each participant. Staff and coordinators were only allowed to contact patients via phone call, email, and REDCap text messaging to inquire about scheduling the participant for their study visit, Actigraph troubleshooting, and Actigraph data collection and upload. Participants who reached out to the coordinators and staff with inquiries related to their Human-DPP or AI-DPP interventions were given the interventions contact information, with no further intervention assistance from the research staff and coordinators.

Description of Sweetch Intervention

The Sweetch app is a fully automated, AI-driven digital diabetes prevention program designed to facilitate behavioral change through personalized coaching, self-tracking, and adaptive interventions. The intervention is grounded in behavioral science, reinforcement learning, and persuasive systems design, utilizing Just-in-Time Adaptive Interventions (JITAI) to optimize participant engagement.

Core Behavioral Change Mechanisms:

Sweetch integrates multiple evidence-based Behavior Change Techniques (BCTs), including:

- Goal setting – Adaptive, AI-driven goal recommendations for physical activity, weight loss, and nutrition.
- Self-monitoring – Real-time tracking of steps, weight, and dietary intake.
- Personalized feedback – AI-generated insights on progress and compliance.
- Contextual recommendations – AI-driven push notifications based on user behavior and environmental data.
- Reinforcement learning – AI dynamically adjusts coaching strategies based on user engagement.
- Gamification elements – Leaderboards, challenges, and achievements for motivation.

AI-Driven Personalization

Sweetch's personalization engine leverages:

1. Historical Data: User demographics, baseline physical activity, weight, and dietary patterns.
2. Real-Time Contextual Data: Location, calendar events, environmental factors (e.g., weather, time of day).
3. Dynamic Behavioral Data: App interactions, step counts, weight fluctuations, goal adherence.
4. AI Adaptation Mechanism: The reinforcement learning algorithm continuously refines recommendations based on user responsiveness, progressively optimizing engagement.

Key Features of the Sweetch App

A. Digital Coaching and Just-in-Time Adaptive Interventions

- AI-generated push notifications sent when participants are most likely to engage.
- Messages tailored to individual goals (e.g., increasing physical activity, reducing calorie intake, completing educational modules).
- Push notifications dynamically adjust based on real-world compliance and user responsiveness.
- Limited to 10 notifications per day (fewer if adherence is high).
- Participants can toggle notifications on/off but cannot modify their timing or frequency.

B. Physical Activity Tracking

- Automatically detects step counts and movement patterns via smartphone accelerometers.
- Manual input option for activities not passively tracked (e.g., swimming, Pilates, cycling).
- AI suggests incremental increases in activity levels based on individual capacity.

C. Weight Tracking

- Bluetooth-enabled body weight scale auto-syncs with the app.
- Option for manual weight entry if not using a digital scale.
- AI provides feedback on weight trends and adaptive goal adjustments.

D. Nutrition Tracking

- Users log meals, snacks, and beverages, with AI-generated calorie balance insights.
- Adaptive dietary recommendations provided based on reported intake.
- Reinforcement learning optimizes dietary interventions by adjusting the complexity of nutritional feedback.

E. T2D Prevention Educational Modules

- Core Phase (Months 1-6): 19 in-app CDC-based diabetes prevention lessons.
- Maintenance Phase (Months 7-12): 6 lessons reinforcing behavioral change.
- Delivery Format: Text, video, and interactive content.

F. Gamification & Engagement Features

- My Achievements: Tracks milestones and provides motivational reinforcement.
- My Challenges: AI-curated challenges to promote habit formation.
- My Competitions & Leaderboard: Social comparison features to foster engagement.

The following describes the Sweetch intervention according to the components of the CONSORT-AI Extension reporting guidelines[8]:

Checklist Item	Details
Version of the AI Algorithm Used	JITAI algorithm 1.0
Acquisition and Selection of Input Data	Data is collected automatically from the user's smartphone and connected devices (Apple Health, Google Health, activity data, location, calendar data--when permissions are enabled). Users can also manually input data, including meal logs, activities not detected by sensors (e.g., swimming), and weight measurements.
Handling of Poor-Quality or Missing Data	Only collected data is used; missing data is not imputed.
Human-AI Interaction in Data Handling	Data is processed exclusively by the AI system. No human intervention is involved in data processing. The AI autonomously selects goal recommendations, educational content, and the timing of notifications.
Output of the AI Intervention	The AI system determines which notifications to send, when to send them, and the content of messages, optimizing for behavior change based on user data.
Role of AI Outputs in Decision-Making	The AI does not engage in clinical decision-making. It provides behavioral nudges, goal-setting recommendations, and habit-forming prompts but does not make clinical determinations.

Sweetch Algorithm Overview

Algorithm Overview

Sweetch is an AI-powered health technology platform designed to predict, prevent, and manage chronic diseases through hyper-personalized, real-time behavioral interventions. It functions as a fully automated digital health coach that dynamically adapts to each user's needs, delivering just-in-time nudges, risk insights, and motivation to support sustained healthy behaviors.

The Sweetch platform includes an adaptive behavioral intervention system that delivers personalized, just-in-time adaptive interventions (JITAIs) encouraging health-promoting behaviors such as increased physical activity and weight loss. Central to this system is a decision-making framework optimized for environments with partial feedback and evolving user states.

The user journey begins with an initial setup phase. During onboarding, Sweetch collects foundational data including the user's health history, conditions, daily routines, and goals. Simultaneously, user profiling engines analyze behavioral and physiological traits, habitual patterns, and contextual information to generate a "digital blueprint"—a persona that forms the basis for personalization. An initial engagement engine uses this data to calibrate early interactions and recommendations.

After the initial setup, the platform enters a continuous adaptation phase driven by Incremental Change Engines (ICEs) and Incremental Change Programs (ICPs), which personalize each user's care journey. The core personalization framework includes a messaging module that tailors content and delivery by analyzing factors like response time, patient profile, and user input—continually optimizing for adherence. The user's digital blueprint is updated dynamically as the system operates in a feedback loop: Sense & Collect Data → Analyze → Respond (Plan, Adapt, Interact). This adaptive cycle refines interventions in real time to deliver the most effective content, features, and goals for each user. User profiling and context engines continuously supply ICEs with updated behavioral and contextual data, allowing the system to evolve with the user's habits, routines, responsibilities, and preferences.

Personalization Framework

Sweetch's adaptive mechanism operates by learning an engagement strategy that maximizes user adherence and behavioral outcomes over time. Each interaction is treated as a decision point where the system evaluates potential actions (e.g., content, tone, timing) and selects the most appropriate one based on the user's current context. This selection process balances testing new options and repeating successful ones.

The core loop involves:

- State recognition: Ingesting contextual information (e.g., activity level, location, time of day)
- Action selection: Choosing an intervention from a predefined set (e.g., motivational message, physical activity prompt)
- Response assessment: Capturing user responses (e.g., engagement, step count increase)
- Strategy update: Adjusting future decisions based on observed outcomes, refining personalization continuously

Algorithm Structure

The personalization engine in the Sweetch system was developed and pre-trained on pooled (cross-user) historical data prior to the trial, but during the trial itself, the deployed algorithm component operated at the individual level only.

Preclinical phase (training):

The model was initially developed using historical data from multiple users (anonymized and aggregated) to estimate general responsiveness patterns to various intervention messages and timing strategies. This approach allowed for learning baseline expected effects of intervention types on physical activity and app engagement.

Deployment phase (trial):

In the live intervention, no cross-user data pooling was used for real-time decision-making. The engine maintained separate state/action histories per participant. Each participant's own past app usage and behavioral activities determined the intervention policy updates for that individual. In other words, the system personalized interventions purely on individual-level feedback once deployed.

Data Sources for Learning and Personalization

The algorithm is powered by multimodal data streams that enable context-aware personalization:

1. User Behavior Data
    a. Interaction with app content.
    b. Activity completion
2. Sensor Data
    a. Physical activity data from smartphones and wearables
    b. Location (e.g. home, work. venues like Gym, Supermarket, Restaurants)
    c. Time of day
3. Health and Demographic Data
    a. Age, sex, clinical diagnoses
    b. Clinical parameters like BMI
    c. Goals achievements

Decision-Making Strategy

The adaptive mechanism relies on three foundational elements:

a. Balancing Exploration and Exploitation

The system includes an explicit strategy for testing new intervention options while prioritizing effective ones. A probabilistic approach was used to occasionally select less optimal messages to support continued learning. This balance shifted over time, with more exploration early and more exploitation as the system gathered user data. Each user followed an independent adaptation path.

b. Response Feedback Loop

The algorithm used a feedback system focused on two primary outcomes:

- Physical activity, measured by active minutes, with diminishing returns to avoid over-promotion
- App engagement, measured by app open in proximity to prompt sending

The reward mechanism placed greater emphasis on physical activity to support health outcomes, while engagement served as a proxy for adherence.

c. Context-Aware Action Selection

Decisions were informed by a rich set of user-specific and situational features, including:

- Temporal (e.g., time of day)
- Spatial (e.g., location)
- Behavioral (e.g., recent activity trends)
- Psychographic (e.g., response to prior interventions)

These variables shaped which actions were selected and when, ensuring high contextual relevance.

Action Options

Interventions varied in:

- Content: motivational tips, reminders, feedback
- Tone: encouraging, directive
- Timing: immediate or scheduled

Definitions of study cohorts

Cohort	Definition
Primary Analysis	All randomized participants
Per Protocol	All randomized participants who completed 12-month study visit and did not use any prohibited medications (steroids, antihyperglycemics, weight loss medications)

Summary of methods for supplementary figures and tables not included in main manuscript

Table/Figure	Population	Description of Method
eFigure 1	–	Screenshots of Personalized Push Notifications and Nutritional Support in Sweetch appintervention were obtained directly from Sweetch Health Ltd.
eFigure 2	–	Screenshots of Connected Health Tracking and Educational Content in the Sweetch App were obtained directly from Sweetch Health Ltd.
eFigure 3	Randomized Population	The forest plot presents age-adjusted risk differences (aRD) (percentage points) with one-sided 95% CIs for the primary and secondary outcomes of diabetes risk reduction at 12 months. The reference line at -15 percentage points represents the non-inferiority margin. The AI-DPP intervention is considered non-inferior to Human-DPP if the one-sided CI does not cross this threshold. Risk difference adjusted for variables imbalanced at baseline (age). Negative aRD values indicate a lower outcome achievement frequency in the AI-DPP group compared to the Human-DPP group.
eFigure 4	Randomized Population	The forest plot presents risk differences (percentage points) with one-sided 95% CIs for the secondary analysis (subgroup analysis by site, sex, age, baseline A1C, and BMI category). The reference line at -15 percentage points represents the non-inferiority margin. The AI-DPP intervention is considered non-inferior to Human-DPP if the one-sided CI does not cross this threshold. Negative risk difference values indicate a lower outcome achievement frequency in the AI-DPP group compared to the Human-DPP group. No multiplicity adjustment was applied as these analyses are exploratory and not intended for formal hypothesis testing.
eTable 1	Randomized Population AI-DPP group	The Sweetch application underwent multiple updates during the study. eTable 1a summarizes each version's release date, key features, and eTable 1b summarizes the number (and percent) of AI-DPP participants who began the program in that version.
eTable 2	Randomized Population Human-DPP group	Participants assigned to the Human-DPP group received their DPP program via synchronous Zoom sessions. Each participant was linked to one of four CDC-recognized DPP sites based on study site affiliation. The table summarizes each site's name, location, CDC recognition status, and number (and percent) of Human-DPP participants assigned.
eTable 3	Randomized Population	Baseline characteristics reported for full randomized cohort using methods as described for Main Manuscript, Table 1.
eTable 4	Randomized Population, Per protocol	Proportions of participants meeting laboratory criteria for DPP eligibility. DPP eligibility determined by: (1) Fasting glucose 100–125 mg/dL within the past year, (2) Study

		A1C 5.7%–6.4% during baseline visit, (3) Clinic A1C 5.7%–6.4% within the past year. Definitions of Study A1C and Clinic A1C are provided in the Methods section of the main manuscript. Chi-squared test used to compare differences in proportions and Wilcoxon rank-sum test used to compare continuous measures by group.
eTable 5	Randomized Population	Baseline characteristics reported by randomization site using methods as described for the Main Manuscript, Table 1.
eTable 6	Randomized Population	Baseline Characteristics by Baseline A1C Status, comparing A1C < 5.7% and 5.7% – 6.4%, using methods as described for Main Manuscript, Table 1.
eTable 7	Randomized Population	Baseline Characteristics by Trial Completion Status, comparing study completers to study drop-outs, using methods as described for Main Manuscript, Table 1.
eTable 8	–	Frequency, handling method, and rationale for handling method of missing data was recorded.
eTable 9	Randomized Population	In-window was defined as completing a 6-month visit within 168–196 days from baseline and a 12-month visit within 351–379 days from baseline. Days outside of window was calculated as the days from the relevant study window to the last day in that window. Mean days outside of window were summarized for participants who were outside of window by assigned group. Chi-squared test was used to compare differences in proportions of participants outside window. Wilcoxon rank-sum test used to compare continuous measures by group.
eTable 10	Randomized Population	Proportion of patients who received prohibited antihyperglycemic, weight loss, or steroid medications by assigned group. GLP-1 receptor agonists were classified as antihyperglycemic agents for the purposes of classification. Information displayed by study group and at the participant level. Chi-squared test used for comparison between study groups.
eTable 10b	Randomized Population	Percentage of participants on prohibited medications by study group.
eTable 11	Per protocol	Baseline characteristics reported for the per protocol cohort by assigned group using methods as described for Main Manuscript, Table 1.
eTable 12	Randomized Population	Participants who achieved the composite outcome were further classified according to which individual component outcomes they met (≥5% weight loss, ≥4% weight loss plus ≥150 minutes/week of physical activity, or ≥0.2% HbA1c reduction). Proportions were calculated as the number of participants achieving each unique combination of outcome components divided by the total number of participants who met the composite outcome, separately for each study arm and overall.
eTable 13	Randomized Population	Participants within the Primary Analysis cohort who had a diabetes-range A1C during the 6-month and/or 12-month

		visit were recorded. These participants were marked having failed to achieve endpoint, regardless of body weight and physical activity improvements. Our study did not adjudicate the diagnosis of diabetes, which requires two confirmatory tests.
eTable 14	Per Protocol	Changes in continuous measures of secondary endpoints (body weight, A1C) and weekly physical activity at 12 months.
eTable 15	Per protocol	Primary outcome and secondary binary outcomes analyzed within the per protocol population using statistical methods described in the Main Manuscript.
eTable 16	Randomized Population	In the primary analysis, participants who did not return for their 12-month visit were classified as treatment failures. As a sensitivity analysis, missing 12-month outcome data were addressed using multiple imputation by chained equations, under the assumption that data were missing at random. The imputation model included study group, baseline values (weight, physical activity, A1C), demographic characteristics, and program engagement metrics (weeks wherein AI-DPP was used for AI-DPP participants and session attendance for Human-DPP participants, normalized within each group). Twenty imputed datasets were generated and combined using Rubin's rules.
eTable 17	Randomized Population	As a sensitivity analysis to assess potential violations of the missing at random assumption, we conducted a pattern mixture model analysis in which outcome achievement rates were differentially imputed for participants with missing 12-month data. Specifically, we assumed that all AI-DPP non-completers failed to achieve the primary outcome (0% success), while Human-DPP non-completers were assumed to have the same outcome achievement rate as observed among completers in that group (37.8%). This scenario imposes a directional bias against the AI-DPP and provides a conservative test of non-inferiority.
eTable 18a	Randomized Population	For the individuals who did not wear their Actigraph monitor during their assigned periods, weekly physical activity was assumed to be 0 in the primary analysis (representing a "worst case" scenario). In this sensitivity analysis of the primary endpoint, we identified individuals who did not meet 100% Actigraph compliance and assumed that they would have achieved 150 weekly min of physical activity had they worn their Actigraph monitor. The table in this section displays the results of this "best-case" scenario sensitivity analysis.
eTable 18b	Randomized Population	Given the limitations of the physical activity measurement described in the main manuscript, we conducted a sensitivity analysis of the primary outcome where we assumed all participants attained the 150 min/week physical activity guideline.

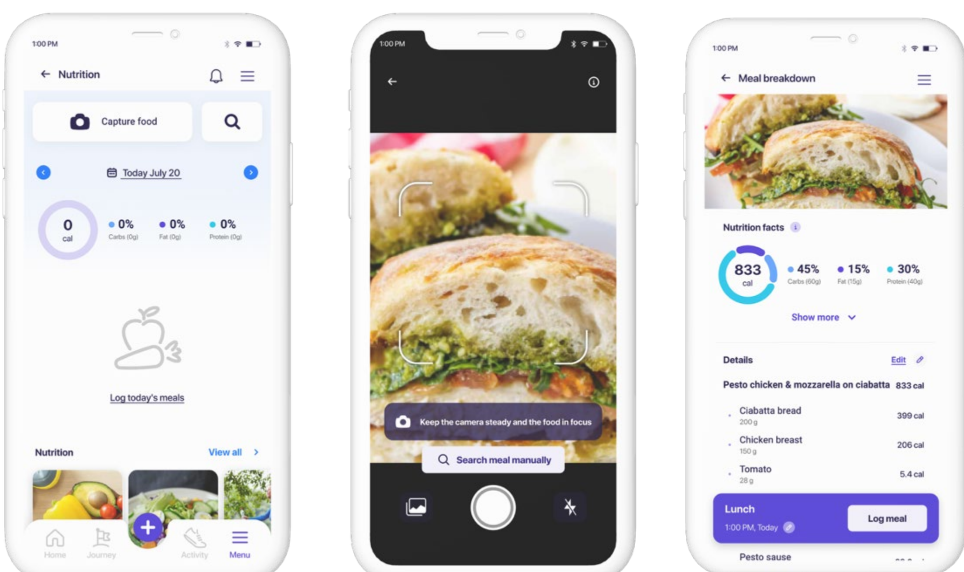
eTable 19a	Randomized Population	Participants in the h-DPP group were referred to one of four distinct DPP programs. While all programs followed the CDC's PreventT2 curriculum and fidelity standards, some heterogeneity in delivery may exist. In this sensitivity analysis, we clustered standard errors by DPP site, treating the four Human-DPPs as distinct clusters (clusters 1–4), and assigned all AI-DPP participants to a fifth cluster.
eTable 19b	Randomized Population	To more granularly approximate potential groupings and shared exposure, we created a variable denoting the site and calendar month of randomization for each participant. This variable (cluster ID) reflects the fact that Human-DPP participants randomized in the same month and at the same DPP site were more likely to be referred to the same cohort, or AI-DPP participants randomized in the same month were more likely to be exposed to the same app version. In this sensitivity analysis, we clustered standard errors by cluster ID.
eTable 18	Randomized Population	Participants within the Primary Analysis cohort who had a diabetes-range A1C during the 6-month and/or 12-month visit were recorded. These participants were marked having failed to achieve endpoint, regardless of body weight and physical activity improvements. Our study did not adjudicate the diagnosis of diabetes, which requires two confirmatory tests.
eTable 19	Per protocol	Changes in continuous measures of secondary endpoints (body weight, A1C) and weekly physical activity at 12 months.
eTable 20a	Randomized Population	Adverse events reported by category, grade, relatedness to the participants' assigned intervention, and by condition. Tables reported number of participants with at least one adverse event by study group, and frequency of adverse events.
eTable 20b	Randomized Population	Adverse events reported by grade. The table reports number of adverse events in each study group.
eTable 20c	Randomized Population	Adverse events reported by relatedness to the participants' assigned intervention. The table reports number of adverse events in each study group.
eTable 20d	Randomized Population	Adverse events reported by condition and grade and assigned intervention.

SUPPLEMENTARY FIGURES AND TABLES

eFigure 1. Screenshots of Personalized Push Notifications and Nutritional Support in Sweetch App

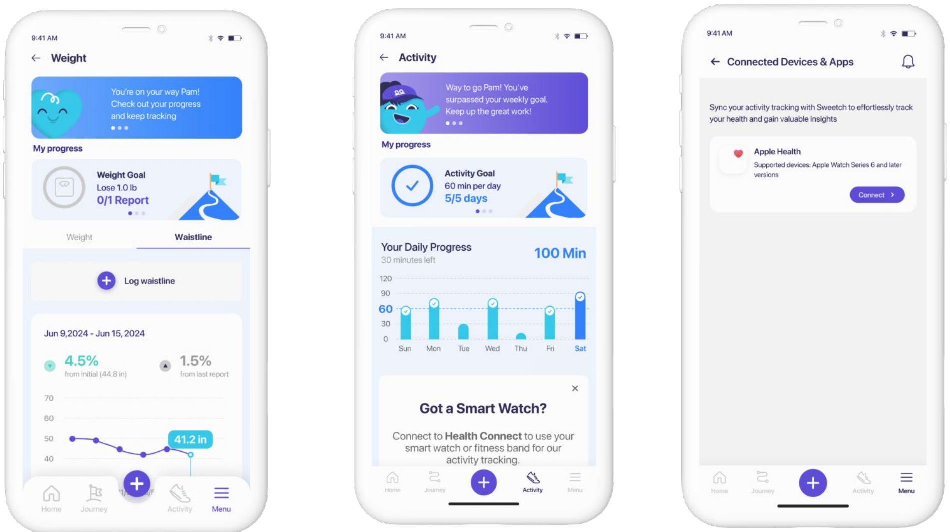
A. Personalized Push Notifications

B. Advanced photo-based meal analysis / Nutrition Tracking

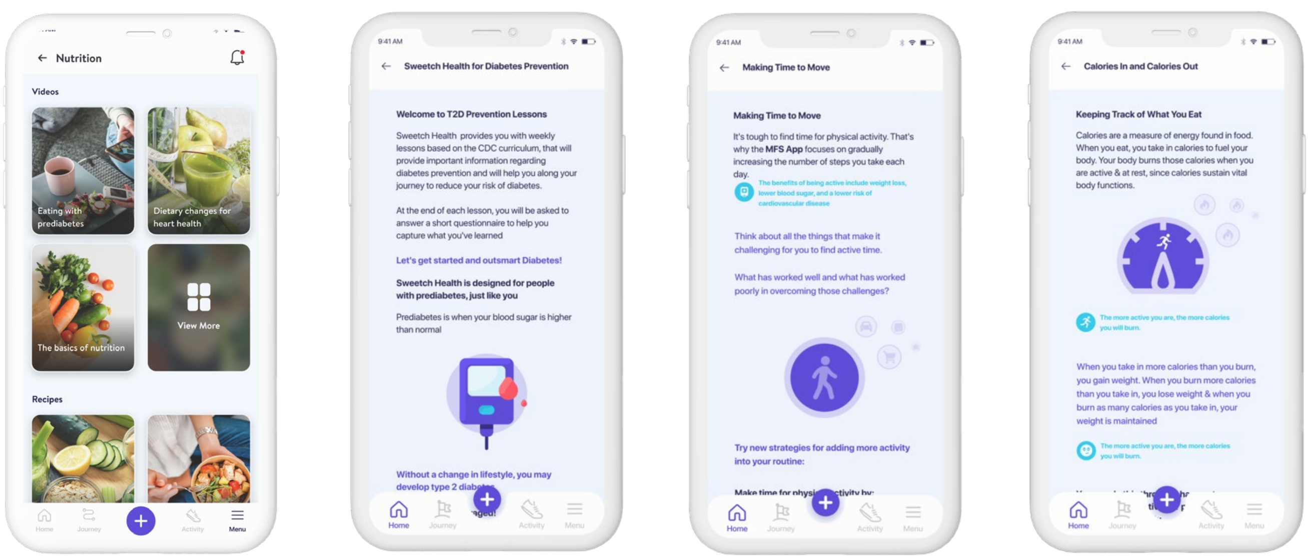


eFigure 2 Screenshots of Connected Health Tracking and Educational Content in Sweetch App

A. Weight and Activity Tracking / Connected Devices

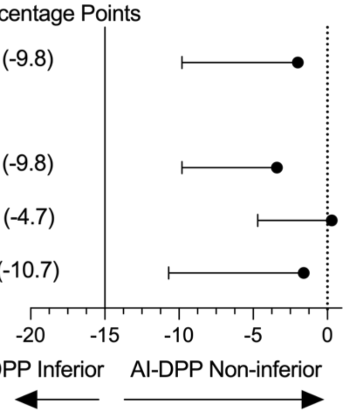


B. Educational Content



eFigure 3. Forest Plot of Adjusted Binary Outcome Differences

Outcome	AI-DPP	Human-DPP	Risk Difference (One-Sided 95% CI)
Primary Analysis[1]	No. of Participants / Total No. (%)		Percentage Points
Diabetes Risk Reduction Benchmark at 12 mo.	58/183 (31.7)	59/185 (31.9)	-2.0 (-9.8)
Elements of Composite Primary Analysis[2]			
5% Weight Loss	31/183 (16.9)	37/185 (20.0)	-3.4 (-9.8)
4% Weight Loss + 150 min. weekly PA	23/183 (12.6)	23/185 (12.4)	0.3 (-4.7)
0.2% A1C Reduction[3]	35/130 (26.9)	35/130 (26.9)	-1.6 (-10.7)



[1]The forest plot presents age-adjusted risk differences (percentage points) with one-sided 95% CIs for the primary and elements of the composite primary analysis at 12 months. The reference line at -15 percentage points represents the non-inferiority margin. The AI-DPP intervention is considered non-inferior to Human-DPP if the one-sided CI does not cross this threshold. Negative RD values indicate a lower outcome achievement frequency in the AI-DPP group compared to the Human-DPP group.

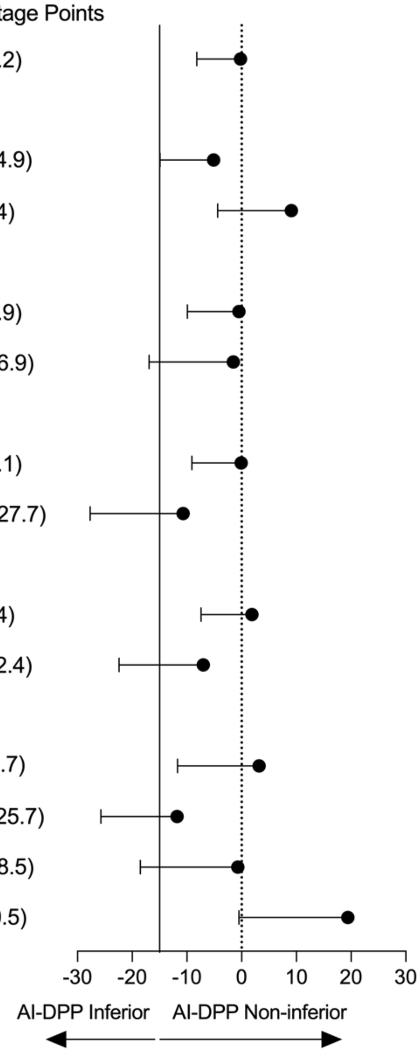
[2]95% confidence intervals for elements of the composite primary analysis outcomes are presented for descriptive purposes only. No multiplicity adjustment was applied as these analyses are exploratory and not intended for formal hypothesis testing.

[3]Analysis of the 0.2% A1C reduction outcome applies only for participants with A1C in the prediabetes range at the study baseline visit (AI-DPP: n=130; H-DPP: n=130)

eFigure 4 Exploratory Subgroup Analysis of Primary Outcome

Diabetes Risk Reduction Outcome at 12 months[1]

Subgroup	AI-DPP	Human-DPP	Risk Difference[2] (One-sided 95% CI[3])
	No. of Participants / Total No. (%)		Percentage Points
Overall Population	58/183 (31.7)	59/185 (31.9)	-0.2 (-8.2)
Site			
Baltimore, MD	36/120 (30.0)	43/123 (35.0)	-5.0 (-14.9)
Reading, Pennsylvania	22/63 (34.9)	16/62 (25.8)	9.1 (-4.4)
Sex			
Female	36/121 (29.8)	42/139 (30.2)	-0.5 (-9.9)
Male	22/62 (35.5)	17/46 (37.0)	-1.5 (-16.9)
Age			
<65 years	32/119 (26.9)	40/148 (27.0)	-0.1 (-9.1)
≥65 years	26/64 (40.6)	19/37 (51.4)	-10.7 (-27.7)
Baseline A1C			
<6.0%	37/123 (30.1)	38/135 (28.1)	1.9 (-7.4)
≥6.0%	21/60 (35.0)	21/50 (42.0)	-7.0 (-22.4)
Baseline BMI Category[4]			
Overweight	17/55 (30.9)	13/47 (27.7)	3.2 (-11.7)
Class I Obesity	20/70 (28.6)	23/57 (40.4)	-11.8 (-25.7)
Class II Obesity	11/33 (33.3)	16/47 (34.0)	-0.7 (-18.5)
Class III Obesity	10/25 (40.0)	7/34 (20.6)	19.4 (-0.5)



[1]The forest plot presents risk differences (percentage points) with one-sided 95% CIs for the primary analysis. The reference line at -15 percentage points represents the non-inferiority margin. The AI-DPP intervention is considered non-inferior to Human-DPP if the one-sided CI does not cross this threshold.

[2]Negative risk difference values indicate a lower outcome achievement frequency in the AI-DPP group compared to the Human-DPP group.

[3]95% confidence intervals for subgroup analyses are presented for descriptive purposes only. No multiplicity adjustment was applied as these analyses are exploratory and not intended for formal hypothesis testing.[4]BMI Classification: Overweight: BMI 25.0–29.9 kg/m²; Class I Obesity: BMI 30.0–34.9 kg/m²; Class II Obesity: BMI 35.0–39.9 kg/m²; Class III Obesity: BMI ≥40.0 kg/m².

eTable 1a. AI-DPP Updates During Trial Period

Version number	Date of Update	Patients Starting with Version (n, % of AI-DPP Arm)	Main Update
122.100.407	December 2024	-	• Re-design of the nutrition domain
11.100.131	September 2023	18 (9.84)	• Home page redesign to display daily navigation of goal achievements
11.100.130	August 2023	3 (1.64)	• Expansion of educational content library
11.100.129	July 2023	8 (4.37)	• Enhancement of physical activity notifications
11.100.126	May 2023	19 (10.38)	• Wearables-based physical activity tracking support • Set default for initial physical activity goal
11.100.110	April 2023	5 (2.73)	• Enhancement of location-based notifications
11.100.106	March 2023	12 (6.56)	• Educational content component added to homepage
11.100.101	August 2022	60 (32.76)	• Sweetch 2.0 - Redesign of app screens - homepage, journey, and app domains (nutrition, physical activity, weight)
Sweetch 1.0	October 2021 Start of trial	58 (31.69)	

Note: See section on Screenshots of Sweetch Intervention for associated visuals for updates.

eTable 1b. AI-DPP Updates During Trial Period

AI Version	Number of Participants Exposed (% of Primary Analysis population assigned to AI-DPP)
Sweetch 1.0 (Start of trial – Oct 2021)	58 (32)
11.100.101 (August 2022)	102 (56)
11.100.106 (March 2023)	109 (60)
11.100.110 (April 2023)	103 (56)
11.100.126 (May 2023)	99 (54)
11.100.129 (July 2023)	105 (57)
11.100.130 (August 2023)	104 (57)
11.100.131 (September 2023)	5 (5)
122.100.407 (December 2024)	5 (5)

eTable 2. Human-DPP Breakdown by Site

DPP Name	Study Recruitment Site	Location of Facility	CDC Recognition Status	Number of Participants Assigned (% of those randomized to Human-DPP)
Brancati Center	JHU	Johns Hopkins Division of General Internal Medicine 2024 E. Monument Street Baltimore, MD 21287	Full Plus	63 (34.1)
University of Maryland Medical Center	JHU	110 S Paca St Suite 5n170, Baltimore, MD 21201-1642	Full Plus	60 (32.4)
Montgomery County	Reading	1430 Dekalb St, Norristown, PA 19401	Full Plus	3 (1.6)
Pottstown Medical Specialists	Reading	1591 Medical Dr. Pottstown, PA 19464	Full Plus	59 (31.9)
Note: All Human-DPP participants received their assigned DPP via distance learning (i.e., synchronous Zoom sessions). JHU = Johns Hopkins University.				

eTable 3. Baseline Characteristics of the Randomized Population

Characteristic[1]	Randomized Population (N=368)
Site — no. (%)	
Johns Hopkins Hospital (Baltimore, MD)	243 (66.0%)
Reading Tower Health (Reading, PA)	125 (34.0%)
Age — yr [2]	58.0 [50.0, 65.0]
Sex — no. (%)	
Female	260 (70.7%)
Male	108 (29.3%)
Race — no. (%)	
American Indian or Alaskan Native	1 (0.3%)
Asian	27 (7.3%)
Black or African American	100 (27.2%)
White or Caucasian	225 (61.1%)
More than One Race	8 (2.2%)
Other	6 (1.6%)
Declined to Answer	1 (0.3%)
Ethnicity — no. (%)	
Not Hispanic or Latino	343 (93.5%)
Hispanic or Latino	20 (5.4%)
Declined to Answer	2 (0.5%)
Unknown	3 (0.8%)
Marital Status — no. (%)	
Married/Partnered	241 (65.5%)
Previously Married (Divorced/Separated/Widowed)	53 (14.4%)
Single/Other	74 (20.1%)
Educational Attainment — no. (%)	
High School or Less	42 (11.4%)
Some College/Associate's Degree	74 (20.1%)
Bachelor's Degree	108 (29.3%)
Graduate/Professional Degree	143 (38.9%)
Declined to Answer	1 (0.3%)
Medical History — no. (%) [3]	
Dyslipidemia	171 (46.5%)
Hypertension	160 (43.5%)
Back pain	96 (26.1%)
Mood disorder	82 (22.3%)
Osteoarthritis	71 (19.3%)
Asthma/COPD	59 (16.0%)
Body Mass Index (BMI) — kg/m²	32.3 [28.5, 37.1]
BMI Classification — no. (%)	
Overweight (BMI 25.0–29.9 kg/m²)	102 (27.7%)
Class I Obesity (BMI 30.0–34.9 kg/m²)	127 (34.5%)
Class II Obesity (BMI 35.0–39.9 kg/m²)	80 (21.7%)
Class III Obesity (BMI ≥40.0 kg/m²)	59 (16.0%)
A1C — % [4]	5.8 ± 0.3
Actigraphy-measured MVPA — min/week [5]	237.0 (105.5, 418.0)

Actigraphy-measured MVPA <150 min/week — no. (%) [6]	119 (32.3%)
Diet Quality — no. (%) [7]	
Healthiest Diet (1-5)	143 (38.9%)
Moderately Healthy Diet (6-10)	108 (29.3%)
Least Healthy Diet (11-16)	117 (31.8%)

Abbreviations: N or no.= number of participants; yr= years; MVPA= moderate-to-vigorous physical activity.

[1]Plus-minus values are means ± SD. Values for non-normally distributed continuous characteristics are presented as median [IQR].

[2]Age differed between study groups ($p = 0.014$); all other baseline characteristics were similar ($p > 0.05$).

[3]Medical history collected via participant self-report at baseline.

[4]A1C reference ranges were defined as <5.7% for normal glycemia, 5.7–6.4% for prediabetes, and ≥6.5% for diabetes.

[5]For participants with no valid baseline Actigraphy data (n=3), MVPA was imputed as 0 min/week to reflect likely inactivity following repeated prompting to wear the device.

[6]Moderate-to-vigorous PA (MVPA) <150 min/week corresponds to not meeting the recommended PA guidelines for prediabetes.

[7]Diet Quality Score: Diet quality at baseline was scored using the Starting the Conversation dietary survey[43], with lower scores indicating healthier dietary patterns. The score ranges from 0 to 16, based on responses to eight items assessing key dietary behaviors (e.g., intake of fast food, fruits, vegetables, sugary drinks, chips, desserts, beans, and margarine or butter), each scored from 0 (most healthful) to 2 (least healthful). For one participant, a missing response imputed with a mid-range value (1 of 0–2) to allow Diet Quality Score calculation.

eTable 4. DPP Eligibility Breakdown

Randomized Population – DPP Eligibility	AI-Based Diabetes Prevention Program (N=183)	Human-Coach-Based Diabetes Prevention Program (N=185)
Study A1C — no. (%) [1]	130 (71.0%)	130 (70.3%)
Study A1C Result — %	5.9 [5.8, 6.1]	5.9 [5.8, 6.1]
Clinic A1C — no. (%)	32 (17.5%)	44 (23.8%)
Clinic A1C Result — %	5.7 [5.7, 5.9]	5.8 [5.7, 5.9]
Fasting Glucose — no. (%) [2]	21 (11.5%)	11 (5.9%)
Fasting Glucose Result — mg/dL	105 [103, 107]	108 [104, 114]
Per-Protocol Population – DPP Eligibility	AI-Based Diabetes Prevention Program (N=151)	Human-Coach-Based Diabetes Prevention Program (N=149)
Study A1C — no. (%)	106 (70.2%)	103 (69.1%)
Study A1C Result — %	5.9 [5.8, 6.1]	5.9 [5.8, 6.0]
Clinic A1C — no. (%)	26 (17.2%)	36 (24.2%)
Clinic A1C Result — %	5.7 [5.7, 5.9]	5.8 [5.7, 5.9]
Fasting Glucose — no. (%)	19 (12.6%)	10 (6.7%)
Fasting Glucose Result — mg/dL	105.0 [103, 112]	106.5 [104, 114]

Note: There were no statistically significant differences between study groups in the Primary Analysis and per-protocol populations regarding DPP eligibility.

[1] A1C reference ranges: normal <5.7%, prediabetes 5.7–6.4%, diabetes ≥6.5%.

[2] Fasting glucose reference ranges: normal <100 mg/dL, impaired fasting glucose 100–125 mg/dL, diabetes ≥126 mg/dL.

eTable 5. Baseline Characteristics by Site

Characteristic[1]	Johns Hopkins Hospital (N=243)	Reading Tower Health (N=125)
Age — yr [2]	57.0 [47.0, 65.0]	61.0 [53.0, 66.0]
Sex — no. (%)		
Female	170 (70.0%)	90 (72.0%)
Male	73 (30.0%)	35 (28.0%)
Race — no. (%)		
American Indian or Alaskan Native	1 (0.4%)	0 (0.0%)
Asian	25 (10.3%)	2 (1.6%)
Black or African American	96 (39.5%)	4 (3.2%)
White or Caucasian	114 (46.9%)	111 (88.8%)
More than One Race	5 (2.1%)	3 (2.4%)
Other	2 (0.8%)	4 (3.2%)
Declined to Answer	0 (0.0%)	1 (0.8%)
Ethnicity — no. (%)		
Not Hispanic or Latino	230 (94.7%)	113 (90.4%)
Hispanic or Latino	11 (4.5%)	9 (7.2%)
Declined to Answer	1 (0.4%)	1 (0.8%)
Unknown	1 (0.4%)	2 (1.6%)
Marital Status — no. (%)		
Married/Partnered	146 (60.1%)	95 (76.0%)
Previously Married (Divorced/Separated/Widowed)	33 (13.6%)	20 (16.0%)
Single/Other	64 (26.3%)	10 (8.0%)
Educational Attainment — no. (%)		
High School or Less	16 (6.6%)	26 (20.8%)
Some College/Associate's Degree	41 (16.9%)	33 (26.4%)
Bachelor's Degree	71 (29.2%)	37 (29.6%)
Graduate/Professional Degree	115 (47.3%)	28 (22.4%)
Declined to Answer	0 (0.0%)	1 (0.8%)
Medical History — no. (%) [3]		
Dyslipidemia	116 (47.7%)	55 (44.0%)
Hypertension	109 (44.9%)	51 (40.8%)
Back Pain	69 (28.4%)	27 (21.6%)
Mood disorder	54 (22.2%)	28 (22.4%)
Osteoarthritis	49 (20.2%)	22 (17.6%)
Asthma/COPD	41 (16.9%)	18 (14.4%)
Body Mass Index (BMI) — kg/m²	32.7 (28.4, 37.4)	32.2 (29.6, 36.8)
BMI Classification — no. (%)		
Overweight (BMI 25.0–29.9 kg/m²)	68 (28.0%)	34 (27.2%)
Class I Obesity (BMI 30.0–34.9 kg/m²)	78 (32.1%)	49 (39.2%)
Class II Obesity (BMI 35.0–39.9 kg/m²)	55 (22.6%)	25 (20.0%)
Class III Obesity (BMI ≥40.0 kg/m²)	42 (17.3%)	17 (13.6%)
A1C — % [4]	5.8 ± 0.3	5.8 ± 0.3
Actigraphy-measured MVPA — min/week [5]	205.0 (84.0, 359.0)	320.0 (160.0, 499.0)

Actigraphy-measured MVPA <150 min/week — no. (%) [6]	90 (37.0%)	29 (23.2%)
Diet Quality — no. (%) [7]		
Healthiest Diet (1-5)	104 (42.8%)	39 (31.2%)
Moderately Healthy Diet (6-10)	69 (28.4%)	39 (31.2%)
Least Healthy Diet (11-16)	70 (28.8%)	47 (37.6%)

Abbreviations: N or no.= number of participants; yr= years; MVPA= moderate-to-vigorous physical activity.

[1]Plus-minus values are means ± SD. Values for non-normally distributed continuous characteristics are presented as median [IQR]. The following baseline characteristics had a statistically significant difference between trial sites: age ($p = 0.017$), race ($p<0.001$), marital status ($p<0.001$), educational attainment ($p<0.001$), and MVPA ($p<0.001$).

[2]Age differed between study groups ($p = 0.014$); all other baseline characteristics were similar ($p > 0.05$).

[3]Medical history collected via participant self-report at baseline.

[4]A1C reference ranges were defined as <5.7% for normal glycemia, 5.7–6.4% for prediabetes, and ≥6.5% for diabetes.

[5]For participants with no valid baseline Actigraphy data (n=3), MVPA was imputed as 0 min/week to reflect likely inactivity following repeated prompting to wear the device.

[6]Moderate-to-vigorous PA (MVPA) <150 min/week corresponds to not meeting the recommended PA guidelines for prediabetes.

[7]Diet Quality Score: Diet quality at baseline was scored using the Starting the Conversation dietary survey[43], with lower scores indicating healthier dietary patterns. The score ranges from 0 to 16, based on responses to eight items assessing key dietary behaviors (e.g., intake of fast food, fruits, vegetables, sugary drinks, chips, desserts, beans, and margarine or butter), each scored from 0 (most healthful) to 2 (least healthful). For one participant, a missing response imputed with a mid-range value (1 of 0–2) to allow Diet Quality Score calculation.

eTable 6. Baseline Characteristics by Baseline A1C Status

Characteristic[1]	Study A1C < 5.7% (N = 108)	Study A1C 5.7 – 6.4% (N = 260)
Site — no. (%)		
Johns Hopkins Hospital (Baltimore, MD)	79 (73.1%)	164 (63.1%)
Reading Tower Health (Reading, PA)	29 (26.9%)	96 (36.9%)
Age — yr [2]	55.5 [46.5, 64.0]	59.0 [51.0, 65.0]
Sex — no. (%)		
Female	75 (69.4%)	185 (71.2%)
Male	33 (30.6%)	75 (28.8%)
Race — no. (%)		
American Indian or Alaskan Native	0 (0.0%)	1 (0.4%)
Asian	6 (5.6%)	21 (8.1%)
Black or African American	32 (29.6%)	68 (26.2%)
White or Caucasian	66 (61.1%)	159 (61.2%)
More than One Race	2 (1.9%)	6 (2.3%)
Other	2 (1.9%)	4 (1.5%)
Declined to Answer	0 (0.0%)	1 (0.4%)
Ethnicity — no. (%)		
Not Hispanic or Latino	96 (88.9%)	247 (95.0%)
Hispanic or Latino	10 (9.3%)	10 (3.8%)
Declined to Answer	0 (0.0%)	2 (0.8%)
Unknown	2 (1.9%)	1 (0.4%)
Marital Status — no. (%)		
Married/Partnered	66 (61.1%)	175 (67.3%)
Previously Married (Divorced/Separated/Widowed)	13 (12.0%)	40 (15.4%)
Single/Other	29 (26.9%)	45 (17.3%)
Educational Attainment — no. (%)		
High School or Less	9 (8.3%)	33 (12.7%)
Some College/Associate's Degree	20 (18.5%)	54 (20.8%)
Bachelor's Degree	30 (27.8%)	78 (30.0%)
Graduate/Professional Degree	49 (45.4%)	94 (36.2%)
Declined to Answer	0 (0.0%)	1 (0.4%)
Medical History — no. (%) [3]		
Dyslipidemia	51 (47.2%)	120 (46.2%)
Hypertension	43 (39.8%)	117 (45.0%)
Back pain	33 (30.6%)	63 (24.2%)
Mood disorder	24 (22.2%)	58 (22.3%)
Osteoarthritis	21 (19.4%)	50 (19.2%)
Asthma/COPD	15 (13.9%)	44 (16.9%)
BMI Classification — no. (%)	32.1 [28.2, 37.0]	32.5 [28.9, 37.2]
Overweight (BMI 25.0–29.9 kg/m²)	32 (29.6%)	70 (26.9%)
Class I Obesity (BMI 30.0–34.9 kg/m²)	41 (38.0%)	86 (33.1%)
Class II Obesity (BMI 35.0–39.9 kg/m²)	18 (16.7%)	62 (23.8%)
Class III Obesity (BMI ≥40.0 kg/m²)	17 (15.7%)	42 (16.2%)

A1C — % [4]	5.5 ± 0.1	5.9 ± 0.2
Actigraphy-measured MVPA — min/week [5]	246.0 [119.5, 451.0]	236.5 [99.5, 415.0]
Actigraphy-measured MVPA <150 min/week — no. (%) [6]	36 (33.3%)	83 (31.9%)
Diet Quality — no. (%) [7]		
Healthiest Diet (1-5)	45 (41.7%)	98 (37.7%)
Moderately Healthy Diet (6-10)	35 (32.4%)	73 (28.1%)
Least Healthy Diet (11-16)	28 (25.9%)	89 (34.2%)

Abbreviations: N or no.= number of participants; yr= years; MVPA= moderate-to-vigorous physical activity.

[1]Plus-minus values are means ± SD. Values for non-normally distributed continuous characteristics are presented as median [IQR]. Site differed between groups ($p = 0.024$). Ethnicity differed between groups ($p = 0.018$). All other baseline characteristics were similar ($p > 0.05$).

[2]Age differed between study groups ($p = 0.014$); all other baseline characteristics were similar ($p > 0.05$).

[3]Medical history collected via participant self-report at baseline.

[4]A1C reference ranges were defined as <5.7% for normal glycemia, 5.7–6.4% for prediabetes, and ≥6.5% for diabetes.

[5]For participants with no valid baseline Actigraphy data (n=3), MVPA was imputed as 0 min/week to reflect likely inactivity following repeated prompting to wear the device.

[6]Moderate-to-vigorous PA (MVPA) <150 min/week corresponds to not meeting the recommended PA guidelines for prediabetes.

[7]Diet Quality Score: Diet quality at baseline was scored using the Starting the Conversation dietary survey[43], with lower scores indicating healthier dietary patterns. The score ranges from 0 to 16, based on responses to eight items assessing key dietary behaviors (e.g., intake of fast food, fruits, vegetables, sugary drinks, chips, desserts, beans, and margarine or butter), each scored from 0 (most healthful) to 2 (least healthful). For one participant, a missing response imputed with a mid-range value (1 of 0–2) to allow Diet Quality Score calculation.

eTable 7. Baseline Characteristics by Trial Completion Status

Characteristic[1]	12-month Completer (N= 313)	Dropped Out or Lost to Follow Up (N=55)
Site — no. (%)		
Johns Hopkins Hospital (Baltimore, MD)	208 (66.5%)	35 (63.6%)
Reading Tower Health (Reading, PA)	105 (33.5%)	20 (36.4%)
Age — yr[2]	59.0 [51.0, 66.0]	55.0 [47.0, 61.0]
Sex — no. (%)		
Female	218 (69.6%)	42 (76.4%)
Male	95 (30.4%)	13 (23.6%)
Race — no. (%)		
American Indian or Alaskan Native	1 (0.3%)	0 (0.0%)
Asian	24 (7.7%)	3 (5.5%)
Black or African American	83 (26.5%)	17 (30.9%)
White or Caucasian	192 (61.3%)	33 (60.0%)
More than One Race	8 (2.6%)	0 (0.0%)
Other	4 (1.3%)	2 (3.6%)
Declined to Answer	1 (0.3%)	0 (0.0%)
Ethnicity — no. (%)		
Not Hispanic or Latino	294 (93.9%)	49 (89.1%)
Hispanic or Latino	15 (4.8%)	5 (9.1%)
Declined to Answer	2 (0.6%)	0 (0.0%)
Unknown	2 (0.6%)	1 (1.8%)
Marital Status — no. (%)		
Married/Partnered	212 (67.7%)	29 (52.7%)
Previously Married (Divorced/Separated/Widowed)	41 (13.1%)	12 (21.8%)
Single/Other	60 (19.2%)	14 (25.5%)
Educational Attainment — no. (%)		
High School or Less	37 (11.8%)	5 (9.1%)
Some College/Associate's Degree	61 (19.5%)	13 (23.6%)
Bachelor's Degree	88 (28.1%)	20 (36.4%)
Graduate/Professional Degree	127 (40.6%)	16 (29.1%)
Declined to Answer	0 (0.0%)	1 (1.8%)
Medical History — no. (%) [3]		
Dyslipidemia	145 (46.3%)	26 (47.3%)
Hypertension	136 (43.5%)	24 (43.6%)
Back pain	85 (27.2%)	11 (20.0%)
Mood disorder	67 (21.4%)	15 (27.3%)
Osteoarthritis	64 (20.4%)	7 (12.7%)
Asthma/COPD	51 (16.3%)	8 (14.5%)
Body Mass Index (BMI) — kg/m²	32.3 [28.5, 37.0]	32.1 [28.6, 37.6]
BMI Classification — no. (%)		
Overweight (BMI 25.0–29.9 kg/m²)	85 (27.2%)	17 (30.9%)
Class I Obesity (BMI 30.0–34.9 kg/m²)	109 (34.8%)	18 (32.7%)
Class II Obesity (BMI 35.0–39.9 kg/m²)	69 (22.0%)	11 (20.0%)
Class III Obesity (BMI ≥40.0 kg/m²)	50 (16.0%)	9 (16.4%)

A1C — % [4]	5.8 ± 0.2	5.8 ± 0.3
Actigraphy-measured MVPA — min/week [5]	240.0 [107.0, 418.0]	191.0 [78.0, 477.0]
Actigraphy-measured MVPA <150 min/week — no. (%) [6]	100 (31.9%)	19 (34.5%)
Diet Quality — no. (%) [7]		
Healthiest Diet (1-5)	128 (40.9%)	15 (27.3%)
Moderately Healthy Diet (6-10)	91 (29.1%)	17 (30.9%)
Least Healthy Diet (11-16)	94 (30.0%)	23 (41.8%)

Abbreviations: N or no.= number of participants; yr= years; MVPA= moderate-to-vigorous physical activity.

[1]Plus-minus values are means ± SD. Values for non-normally distributed continuous characteristics are presented as median [IQR]. No baseline characteristics were statistically significant different between groups ($p<0.05$).

[2]Age differed between study groups ($p = 0.014$); all other baseline characteristics were similar ($p > 0.05$).

[3]Medical history collected via participant self-report at baseline.

[4]A1C reference ranges were defined as <5.7% for normal glycemia, 5.7–6.4% for prediabetes, and ≥6.5% for diabetes.

[5]For participants with no valid baseline Actigraphy data (n=3), MVPA was imputed as 0 min/week to reflect likely inactivity following repeated prompting to wear the device.

[6]Moderate-to-vigorous PA (MVPA) <150 min/week corresponds to not meeting the recommended PA guidelines for prediabetes.

[7]Diet Quality Score: Diet quality at baseline was scored using the Starting the Conversation dietary survey[43], with lower scores indicating healthier dietary patterns. The score ranges from 0 to 16, based on responses to eight items assessing key dietary behaviors (e.g., intake of fast food, fruits, vegetables, sugary drinks, chips, desserts, beans, and margarine or butter), each scored from 0 (most healthful) to 2 (least healthful). For one participant, a missing response imputed with a mid-range value (1 of 0–2) to allow Diet Quality Score calculation.

eTable 8a. Missing Data: Survey Data

Variable	Missing Instances		Method for Handling[1]
	AI-Based Diabetes Prevention Program (N=183) (%)	Human-Coach-Based Diabetes Prevention Program (N=185)	
Diet Quality Score	1 (0.6)	0 (0)	Missing answer to question was assigned middle value of 1 (0-2)
[1]Rationale: 1 survey question out of 8 questions was missing, preventing calculating of diet quality score. Imputation avoided extremes. Remaining 7 values used to calculate diet quality score.			

eTable 8b. Missing Data: Actigraphy Data

Variable	**Missing Instances**		**Method for Handling**[1]
	AI-Based Diabetes Prevention Program (N=183) (%)	**Human-Coach-Based Diabetes Prevention Program (N=185) (%)**	
Physical Activity Assessment			
Baseline	8 (4.4)	5 (2.7)	Assigned 0 minutes/week
Month 1	17 (9.3)	21 (11.4)	Assigned 0 minutes/week to periods of non-wear
Month 2	22 (12)	35 (18.9)	
Month 3	23 (12.6)	31 (16.8)	
Month 4	27 (14.8)	35 (18.9)	
Month 5	43 (23.5)	38 (20.5)	
Month 6	40 (21.9)	41 (22.2)	
Month 7	49 (26.8)	54 (29.2)	
Month 8	48 (26.2)	54 (29.2)	
Month 9	48 (26.2)	54 (29.2)	
Month 10	52 (28.4)	62 (33.5)	
Month 11	60 (32.8)	65 (35.1)	

[1]Rationale: Participants were instructed to wear an ActiGraph device for one week each month to measure moderate-to-vigorous physical activity. Each month, study staff sent reminders and follow-up messages to prompt device use. If no valid wear data were recorded despite this outreach, MVPA was imputed as zero for that month to avoid overestimating physical activity. This approach assumes that failure to wear the monitor after repeated prompting may reflect disengagement and low likelihood of physical activity. Given that activity was assessed monthly over a year, this method minimizes the impact of any single missing week and provides a reasonable approximation of long-term behavior. Actigraphy non-compliance did not alter a participant's ability to achieve the primary endpoint through A1C or body weight change.

eTable 8c. Missing Data: 12-Month Measures

Variable	Missing Instances		Method for Handling[1]
	AI-Based Diabetes Prevention Program (N=183) (%)	**Human-Coach-Based Diabetes Prevention Program (N=185) (%)**	
A1C at 12 months	26 (14.2)	29 (15.7)	• Assuming Treatment Failure • Imputation Sensitivity Analysis
Body Weight at 12 months	26 (14.2)	29 (15.7)	• Assuming Treatment Failure • Imputation Sensitivity Analysis

[1] Rationale: There was no missing A1C or body weight data among study completers. Missingness occurred exclusively among participants who dropped out or were lost to follow-up. In the primary analysis, these participants were assumed not to have met the clinical endpoint. In the sensitivity analysis, outcomes were imputed using multiple imputation by chained equations.

eTable 9. Adherence to Follow-Up Schedule

Result	6-Month Visit		12-Month Visit	
	AI-Based Diabetes Prevention Program	**Human-Coach-Based Diabetes Prevention Program**	**AI-Based Diabetes Prevention Program**	**Human-Coach-Based Diabetes Prevention Program**
No. of Attendees / Total (%)	164 / 183 (89.6%)	161 / 185 (87.0%)	157 / 183 (85.8%)	156 / 185 (84.3%)
Mean Days on Study at Visit	184.92 (SD 13.44)	183.84 (SD 7.76)	366.78 (SD 9.37)	367.42 (SD 7.79)
No. of Attendees with Visit Outside of Window[1] / Total (%)	10 / 164 (6.1%)	10 / 161 (6.2%)	7 / 157 (4.5%)	12 / 156 (7.7%)
Mean Days Outside of Window[2]	27.4 (SD 36.25)	12.1 (SD 6.37)	20.14 (SD 13.17)	6.67 (SD 5.18)

Abbreviations: No.= number; SD= standard deviation.

[1]In-window was defined as completing a 6-month visit within 168–196 days from baseline and a 12-month visit within 351–379 days from baseline.

[2]Mean days outside the 12-month visit window differed significantly between groups ($p = 0.016$). All other differences in the table were not statistically significant ($p>0.05$).

eTable 10a. Participants on Prohibited Medications

Instance	Study Arm	Type of Prohibited Medication	Medication
Instance 1	Human	Antihyperglycemic	Semaglutide
Instance 2	Human	Antihyperglycemic	Metformin
Instance 3	Human	Steroid	Methylprednisolone
Instance 4	Human	Weight Loss	Naltrexone
Instance 5	Human	Weight Loss	Phentermine
Instance 6	Human	Antihyperglycemic	Ozempic
Instance 7	Human	Weight Loss	Phentermine
Instance 8	AI	Antihyperglycemic	Semaglutide
Instance 9	AI	Steroid, Antihyperglycemic	Prednisone, Metformin, Insulin
Instance 10	AI	Antihyperglycemic	Semaglutide
Instance 11	AI	Antihyperglycemic	Metformin
Instance 12	AI	Antihyperglycemic	Metformin
Instance 13	AI	Antihyperglycemic	Metformin

eTable 10b. Prohibited Medication

Result	**AI-Based Diabetes Prevention Program** (N=183) (%)	**Human-Coach-Based Diabetes Prevention Program** (N=185) (%)	p-value[1]
Participants on Prohibited Medications	6 (3.3)	7 (3.8)	0.793
[1]Wilcoxon Rank Sum Test			

eTable 11. Baseline Characteristics for participants with 12-month outcome data who did not initiate prohibited medications

Characteristic[1]	**AI-Based Diabetes Prevention Program** (N=151)	**Human-Coach-Based Diabetes Prevention Program** (N=149)
Site — no. (%)		
Johns Hopkins Hospital (Baltimore, MD)	93 (61.6%)	103 (69.1%)
Reading Tower Health (Reading, PA)	58 (38.4%)	46 (30.9%)
Age — yr [2]	61.0 [52.0, 69.0]	59.0 [50.0, 63.0]
Sex — no. (%)		
Female	96 (63.6%)	111 (74.5%)
Male	55 (36.4%)	38 (25.5%)
Race — no. (%)		
American Indian or Alaskan Native	0 (0.0%)	1 (0.7%)
Asian	10 (6.6%)	13 (8.7%)
Black or African American	35 (23.2%)	39 (26.2%)
White or Caucasian	99 (65.6%)	90 (60.4%)
More than One Race	3 (2.0%)	5 (3.4%)
Other	4 (2.6%)	0 (0.0%)
Declined to Answer	0 (0.0%)	1 (0.7%)
Ethnicity — no. (%)		
Not Hispanic or Latino	146 (96.7%)	135 (90.6%)
Hispanic or Latino	3 (2.0%)	12 (8.1%)
Declined to Answer	1 (0.7%)	1 (0.7%)
Unknown	1 (0.7%)	1 (0.7%)
Marital Status — no. (%)		
Married/Partnered	102 (67.5%)	102 (68.5%)
Previously Married (Divorced/Separated/Widowed)	23 (15.2%)	17 (11.4%)
Single/Other	26 (17.2%)	30 (20.1%)
Educational Attainment — no. (%)		
High School or Less	22 (14.6%)	15 (10.1%)
Some College/Associate's Degree	31 (20.5%)	26 (17.4%)
Bachelor's Degree	39 (25.8%)	43 (28.9%)
Graduate/Professional Degree	59 (39.1%)	65 (43.6%)
Medical History — no. (%) [3]		
Dyslipidemia	67 (44.4%)	73 (49.0%)
Hypertension	67 (44.4%)	65 (43.6%)
Back pain	44 (29.1%)	38 (25.5%)
Mood disorder	31 (20.5%)	33 (22.1%)
Osteoarthritis	34 (22.5%)	28 (18.8%)
Asthma/COPD	26 (17.2%)	23 (15.4%)
Body Mass Index (BMI) — kg/m²	32.3 [28.2, 35.9]	32.2 [29.1, 37.6]
BMI Classification — no. (%)		
Overweight (BMI 25.0–29.9 kg/m²)	44 (29.1%)	39 (26.2%)
Class I Obesity (BMI 30.0–34.9 kg/m²)	59 (39.1%)	47 (31.5%)

Class II Obesity (BMI 35.0–39.9 kg/m²)	28 (18.5%)	38 (25.5%)
Class III Obesity (BMI ≥40.0 kg/m²)	20 (13.2%)	25 (16.8%)
A1C — % [4]	5.8 ± 0.3	5.8 ± 0.3
Actigraphy-measured MVPA — min/week [5]	250.0 [112.0, 404.0]	236.0 [104.0, 432.0]
Actigraphy-measured MVPA <150 min/week — no. (%) [6]	42 (27.8%)	52 (34.9%)
Diet Quality — no. (%) [7]		
Healthiest Diet (1-5)	58 (38.4%)	64 (43.0%)
Moderately Healthy Diet (6-10)	42 (27.8%)	45 (30.2%)
Least Healthy Diet (11-16)	51 (33.8%)	40 (26.8%)

Abbreviations: N or no.= number of participants; yr= years; MVPA= moderate-to-vigorous physical activity.

[1]Plus-minus values are means ± SD. Values for non-normally distributed continuous characteristics are presented as median [IQR]. Age differed between study groups ($p = 0.010$). Sex differed between study groups ($p = 0.041$). All other baseline characteristics were similar ($p > 0.05$).

[2]Age differed between study groups ($p = 0.014$); all other baseline characteristics were similar ($p > 0.05$).

[3]Medical history collected via participant self-report at baseline.

[4]A1C reference ranges were defined as <5.7% for normal glycemia, 5.7–6.4% for prediabetes, and ≥6.5% for diabetes.

[5]For participants with no valid baseline Actigraphy data (n=3), MVPA was imputed as 0 min/week to reflect likely inactivity following repeated prompting to wear the device.

[6]Moderate-to-vigorous PA (MVPA) <150 min/week corresponds to not meeting the recommended PA guidelines for prediabetes.

[7]Diet Quality Score: Diet quality at baseline was scored using the Starting the Conversation dietary survey[43], with lower scores indicating healthier dietary patterns. The score ranges from 0 to 16, based on responses to eight items assessing key dietary behaviors (e.g., intake of fast food, fruits, vegs, sugary drinks, chips, desserts, beans, and margarine or butter), each scored from 0 (most healthful) to 2 (least healthful). For one participant, a missing response imputed with a mid-range value (1 of 0–2) to allow Diet Quality Score calculation.

eTable 12 Distribution of Outcomes Achieved Among Participants Who Met Composite Outcome

Individual Components of Composite Outcome			**Proportion Achieving Outcome(s), N (%)**[a]		
≥5% Weight Loss	**≥4% Weight Loss + ≥150 min/week of Physical Activity**	**≥0.2% A1C Reduction**	**AI-led DPP N = 58**	**Human-led DPP N = 59**	**Overall N = 117**
✓	✓	✓	5 (9%)	10 (17%)	15 (13%)
✓	✓		14 (24%)	10 (19%)	24 (21%)
✓		✓	6 (10%)	5 (8%)	11 (9%)
	✓	✓	1 (2%)	1 (2%)	2 (2%)
✓			6 (10%)	12 (20%)	18 (15%)
	✓		3 (5%)	2 (3%)	5 (4%)
		✓	23 (40%)	19 (32%)	42 (36%)

Abbreviations: N= number of participants.

[a]Percentages are calculated out of participants who achieved the composite outcome within each study group and overall.

eTable 13. Participants with Diabetes-Range A1C

Participant*	Study Group	A1C Baseline, %	A1C 6 Months, %	A1C 12 Months, %
Instance 1	Human-DPP	6	**6.7**	6.3
Instance 2	Human-DPP	5.5	**6.9**	**7.9**
Instance 3	Human-DPP	5.9	5.6	**6.8**
Instance 4	Human-DPP	6.3	**6.5**	6.2
Instance 5	Human-DPP	6.3	**6.6**	**6.6**
Instance 6	Human-DPP	5.6	6	**6.7**
Instance 7	Human-DPP	6.1	**6.5**	6.3
Instance 8	AI-DPP	5.9	**7**	5.9
Instance 9	AI-DPP	6.4	**6.5**	6
Instance 10	AI-DPP	6.2	6	**6.5**
Instance 11	AI-DPP	6.4	**6.5**	**6.5**
Instance 12	AI-DPP	6.4	**6.5**	**6.5**
Instance 13	AI-DPP	5.9	6.1	**7.1**
Instance 14	AI-DPP	6.1	6.2	**10.9**
Instance 15	AI-DPP	6.3	6.3	**13.8**
Bolded values indicate A1C ≥ 6.5% (i.e. diabetes range). Note: Study team did not adjudicate the diagnosis of diabetes, which requires two confirmatory tests.				

eTable 14 Continuous Outcomes

Outcome	AI-Based Diabetes Prevention Program	N	Human-Coach-Based Diabetes Prevention Program	N
Percentage Change in Weight (Baseline to 12 Months) (%)	-1.03 [-3.73, 1.46]	151	-1.43 [-4.03, 1.14]	149
Change in Weight (Baseline to 12 Months) (kg)	-1.00 [-3.20, 1.50]	151	-1.30 [-4.30, 1.10]	149
Change in A1C (Baseline to 12 Months) (%)	0.0 [-0.2, 0.1]	106	-0.1 [-0.2, 0.1]	103
Average Physical Activity (min/week)	210.5 [134.5, 344]	151	174.4 [73.9, 321.25]	149
Values for non-normally distributed continuous characteristics are presented as median [IQR]. Values pertain to population with complete data at 12 months who did not take prohibited medications.				

eTable 15. Analysis Restricted to Participants with 12-Month Data Who Did Not Initiate Prohibited Medications

Outcome	AI-Based Diabetes Prevention Program	Human-Coach-Based Diabetes Prevention Program	Risk Difference (One-Sided 95% CI)
	no. of patients/total no. (%)		percentage points
Primary Outcome: Diabetes Risk Reduction Benchmark at 12 mo.	54/151 (35.8)	53/149 (35.6)	0.2 (-8.9)
Secondary Outcomes at 12 mo.			
5% Weight Loss	28/151 (18.5)	33/149 (22.1)	-3.6 (-11.3)
4% Weight Loss + 150 Min. of Weekly Physical Activity	22/151 (14.6)	22/149 (14.8)	-0.2 (-6.9)
0.2% A1C Reduction[a]	33/106 (31.1)	31/103 (30.1)	1.0 (-9.5)

[a]A1C metrics pertain to the population with an A1C in the prediabetes range at the study baseline visit.

Note: The per-protocol analysis was intended as a supportive analysis limited to participants with complete outcome data and without pharmacologic confounding. It does not estimate the causal effect of adherence to a specific treatment strategy. Given the pragmatic design and absence of enforced post-referral intervention protocol, methods such as inverse probability of censoring weighting or the g-formula were not applicable.

eTable 16. Sensitivity Analysis Imputing Outcomes in Those with Missing 12 Month Outcomes Data Using Multiple Imputation by Chain Equations

Outcome	**AI-Based Diabetes Prevention Program**	**Human-Coach-Based Diabetes Prevention Program**	**Risk Difference (One-Sided 95% CI)**
	% achieving outcome (MI-pooled[1])		percentage points
Primary Outcome: Diabetes Risk Reduction at 12 mo. (Pooled)	32.2	31.9	-1.1 (-11.5)

[1]MI= multiple imputation

eTable 17. Sensitivity Analysis of Primary Endpoint: Pattern Mixture Model Assuming Differential Outcome Rates Among Non-Completers

Outcome	**AI-Based Diabetes Prevention Program**	**Human-Coach-Based Diabetes Prevention Program**	**Risk Difference (One-Sided 95% CI)**
	no. of patients/total no. (%)		percentage points
Primary Outcome: Diabetes Risk Reduction at 12 mo.[1]	58/183 (31.7)	70/185 (37.8%)	−6.1 (−14.3)

[1]Assuming 0% success in AI-DPP non-completers, 37.8% in Human-DPP non-completers

eTable 18a. Sensitivity Analysis of Primary Endpoint: “Best-Case” Scenario for Actigraph Non-Wear Periods

Outcome	AI-Based Diabetes Prevention Program	Human-Coach-Based Diabetes Prevention Program	Risk Difference (One-Sided 95% CI)
	no. of patients/total no. (%)		percentage points
Primary Outcome: Diabetes Risk Reduction at 12 mo.[1]	58/183 (31.7)	60/185 (32.4)	-0.74 (-8.8)

[1]Best-Case Scenario assumed that individuals who did not meet actigraphy compliance would have achieved 150 minutes of weekly physical activity had they worn the device.

eTable 18b. Sensitivity Analysis of Primary Endpoint: Assuming Attainment of Physical Activity Goal for all participants

Outcome	AI-Based Diabetes Prevention Program	Human-Coach-Based Diabetes Prevention Program	Risk Difference (One-Sided 95% CI)
	no. of patients/total no. (%)		percentage points
Primary Outcome: Diabetes Risk Reduction at 12 mo.[1]	58/183 (31.7)	60/185 (32.4)	-0.74 (-8.8)

[1]Assumed that all participants attained the 150 minute/week physical activity guideline.

eTable 19a Sensitivity Analysis Using Cluster-Robust Standard Errors by DPP Site

Outcome	**AI-Based Diabetes Prevention Program**	**Human-Coach-Based Diabetes Prevention Program**	**Risk Difference (One-Sided 95% CI)**
	no. of patients/total no. (%)		percentage points
Primary Outcome: Diabetes Risk Reduction at 12 mo.[1]	58/183 (31.7)	59/185 (31.9)	-0.20 (-4.8)

[1]Clustered standard errors by DPP site, treating the four Human-DPPs as distinct clusters (1-4) and assigned all AI-DPPs to a fifth cluster.

eTable 19b Sensitivity Analysis Using Cluster-Robust Standard Errors by Site and Month of Randomization

Outcome	**AI-Based Diabetes Prevention Program**	**Human-Coach-Based Diabetes Prevention Program**	**Risk Difference (One-Sided 95% CI)**
	no. of patients/total no. (%)		percentage points
Primary Outcome: Diabetes Risk Reduction at 12 mo.[1]	58/183 (31.7)	59/185 (31.9)	-0.20 (-6.8)

[1]Clustered participants into cohorts by DPP site (four Human-DPP clusters, 1 AI-DPP cluster) and month of randomization.

eTable 20a. Adverse Events by Category

	AI-Based Diabetes Prevention Program (N=183)	Human-Coach-Based Diabetes Prevention Program (N=185)
# of participants with ≥1 event	66/183 (36.1%)[1]	21/185 (11.4%)
Adverse Event Categories	**No. of AEs (% of events)[2]**	
Musculoskeletal and connective tissue disorders	15 (15.0%)	5 (20.0%)
Infections and infestations	14 (14.0%)	6 (24.0%)
Gastrointestinal disorders	11 (11.0%)	2 (8.0%)
Nervous system disorders	11 (11.0%)	0 (0.0%)
General disorders and administration site conditions	8 (8.0%)	2 (8.0%)
Respiratory, thoracic, and mediastinal disorders	7 (7.0%)	1 (4.0%)
Injury, poisoning, and procedural complications	6 (6.0%)	2 (8.0%)
Renal and urinary disorders	6 (6.0%)	1 (4.0%)
Cardiac disorders	5 (5.0%)	2 (8.0%)
Neoplasms (benign, malignant, and unspecified)	4 (4.0%)	1 (4.0%)
Surgical and medical procedures	4 (4.0%)	1 (4.0%)
Immune system disorders	3 (3.0%)	1 (4.0%)
Endocrine disorders	3 (3.0%)	0 (0.0%)
Hematologic disorders	2 (2.0%)	0 (0.0%)
Psychiatric disorders	1 (1.0%)	1 (4.0%)

[1] Counts and percentages reported at the participant level

[2] Counts and percentages reported at the event level

eTable 20b. Adverse Events by Grade

Adverse Event Grade	**AI-Based Diabetes Prevention Program**	**Human-Coach-Based Diabetes Prevention Program**
Grade 1 (mild)	13	5
Grade 2 (moderate)	42	11
Grade 3 (severe)	43	8
Grade 4 (life-threatening)	2	1
Total No. of Events	100	25

eTable 20c. Adverse Events by Relatedness to Assigned Intervention

Adverse Event Relatedness	**AI-Based Diabetes Prevention Program**	**Human-Coach-Based Diabetes Prevention Program**
Definitely Related	0	0
Probably Related	0	0
Possibly Related	0	0
Not Related	100	25

eTable 20d. Adverse Events by Condition and Grade

Adverse Event	Number of Adverse Events	
	AI-Based Diabetes Prevention Program	**Human-Coach-Based Diabetes Prevention Program**
Grade 1	**13**	**5**
COVID-19	5	4
Exacerbation of asthma	1	
Facial laceration	1	
Generalized anxiety disorder	1	
Hypertension		1
Influenza	1	
Osteopenia	1	
Plantar fasciitis	1	
Sinusitis	1	
Trigger finger	1	
Grade 2	**42**	**11**
Abdominal pain	1	
Anemia	1	
Anxiety		1
Back pain	1	1
Bacterial vaginosis	1	
Benign prostatic hyperplasia		1
Bleeding postoperative		1
Bronchitis		1
COVID-19	2	
Chest pain	1	1
Concussion	1	
Dermatitis allergic	3	
Diarrhea	1	
Diverticulitis	1	
Eye injury		1
Foreign body, ear	1	
Fracture of the distal radius	1	
Gallbladder polyps	1	
Gastroparesis	1	
Gout	1	
Heel spur	1	
Hip injury	1	
Hypercalcemia	1	
Hypercholesterolemia	1	
Hypertension	1	
Inflammatory bowel disease	1	
Influenza		1
Leg pain		1
Lipoma excision		1
Lumbar vertebral fracture	1	
Meniscus tear	1	1

Migraine	2	
Nasal polyp	1	
Nephrolithiasis	2	
Neuralgia	1	
Neuropathy	1	
Neutropenia	1	
Prostate cancer	1	
Sialoadenitis	1	
Sinusitis	2	
Sleep apnea	1	
Urinary tract infection	2	
Vomiting	1	
Grade 3	**43**	**8**
AV block	1	
Allergic reaction		1
Ankle fracture	1	
Atrial fibrillation	3	
Back pain		1
Brain tumor	1	
Breast cancer	2	1
COVID-19	4	1
Chest pain	3	
Diabetes	1	
Diarrhea		1
Ear pain	1	
Flank pain	3	
Foot drop	1	
Foot surgery	1	
Hammer toe surgery	1	
Headache	1	
Hip fracture		1
Hip pain	1	
Inflammatory bowel disease	1	
Interstitial lung disease	1	
Knee pain		1
Melanoma	1	
Meniscus tear	1	
Nephrolithiasis	1	
Pancreatitis	1	
Parkinson's disease	1	
Prostate Cancer	1	
Rectal bleeding	1	
Rotator cuff surgery	1	
Stem cell procedure	1	
Supraventricular tachycardia		1
Surgical and medical procedures	2	
Syncope	1	
Temporal arteritis	1	

Vasovagal syncope	1	
Vertigo	1	
Wrist surgery	1	
Grade 4	**2**	**1**
Appendicitis	1	
Perforated gallbladder		1
Pulmonary embolism	1	

REFERENCES

1. Tuomilehto J, Lindstrom J, Eriksson JG, et al. Prevention of type 2 diabetes mellitus by changes in lifestyle among subjects with impaired glucose tolerance. N Engl J Med 2001;344(18):1343-50. DOI: 10.1056/NEJM200105033441801.

2. Knowler WC, Barrett-Connor E, Fowler SE, et al. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. N Engl J Med 2002;346(6):393-403. DOI: 10.1056/NEJMoa012512.

3. Hamman RF, Wing RR, Edelstein SL, et al. Effect of weight loss with lifestyle intervention on risk of diabetes. Diabetes care 2006;29(9):2102-7. DOI: 10.2337/dc06-0560.

4. Boonpor J, Parra-Soto S, Petermann-Rocha F, et al. Dose-response relationship between device-measured physical activity and incident type 2 diabetes: findings from the UK Biobank prospective cohort study. BMC Med 2023;21(1):191. DOI: 10.1186/s12916-023-02851-5.

5. Lindstrom J, Louheranta A, Mannelin M, et al. The Finnish Diabetes Prevention Study (DPS): Lifestyle intervention and 3-year results on diet and physical activity. Diabetes care 2003;26(12):3230-6. DOI: 10.2337/diacare.26.12.3230.

6. Montoye AHK, Clevenger KA, Pfeiffer KA, et al. Development of cut-points for determining activity intensity from a wrist-worn ActiGraph accelerometer in free-living adults. J Sports Sci 2020;38(22):2569-2578. DOI: 10.1080/02640414.2020.1794244.

7. Liu F, Wanigatunga AA, Schrack JA. Assessment of Physical Activity in Adults Using Wrist Accelerometers. Epidemiol Rev 2022;43(1):65-93. DOI: 10.1093/epirev/mxab004.

8. Liu X, Cruz Rivera S, Moher D, et al. Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension. Nat Med 2020;26(9):1364-1374. DOI: 10.1038/s41591-020-1034-x.